

Fasciola

**Background document for the
WHO Guidelines for drinking-water quality and
the WHO Guidelines on sanitation and health**



**World Health
Organization**

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Highlights

- There is moderate evidence linking transmission of *Fasciola* with unsafe drinking-water and inadequate sanitation. Fascioliasis transmission through ingestion of contaminated surface water is supported by indirect evidence, including statistically significant associations of fascioliasis and other waterborne parasitic infections.
- *Fasciola* are vector-borne zoonotic pathogens. Sheep, goats, cattle, buffalo, pigs and equines are the primary hosts of *Fasciola*. Humans can also be hosts.
- *Fasciola* have a two-host lifecycle involving release of eggs from infected hosts, passage through lymnaeid snails and shedding of cercariae which transform into infectious metacercariae upon attachment to freshwater plants. Consumption of uncooked freshwater plants (e.g. watercress) is the main route of transmission, but contaminated water may also be important. Person-to-person transmission is not possible.
- Asymptomatic infection occurs in 12% to 15% of cases. The acute phase of symptomatic fascioliasis lasts for 3 to 4 months and includes fever, abdominal pain, gastrointestinal disturbance, urticarial and respiratory symptoms. The chronic phase can last from 9 to 14 years, with symptoms including fatty food intolerance, nausea, jaundice, pruritus, right upper-quadrant abdominal tenderness and biliary colic.
- In endemic areas, prevalence peaks in 9- to 11-year-olds; in developed countries, prevalence peaks in 30- to 40-year olds. Fascioliasis follows seasonal patterns influenced by temperature and rainfall, which vary depending on latitude and altitude and can include year-long transmission, mono-seasonal transmission and bi-seasonal transmission.
- Evidence indicates that metacercariae are highly infective, but the infective dose has not been determined.
- Microscopic methods are available to detect and identify metacercariae in faeces, wastewater, on plants and in water. Molecular methods are also available.
- *Fasciola* are found in freshwaters inhabited by lymnaeid snails. Metacercariae can survive for long periods of time in water.
- Systematic approaches that identify and control faecal contamination along the whole sanitation service and drinking-water supply chains are needed to manage microbial risks, including from *Fasciola*. Since animals are the primary source of *Fasciola*, a comprehensive approach that also deals with the management of animal faeces is necessary to control transmission.
- Reduction of *Fasciola* by wastewater and drinking-water treatment varies depending on the process. Removal is based on physical removal (e.g. size exclusion). *Fasciola* metacercariae are likely to be resistant to disinfection processes such as chlorination.
- Due to the complex life cycle and environmental survival of *Fasciola*, *E. coli* (or, alternatively, thermotolerant coliforms) is not a suitable indicator for the possible presence or absence of *Fasciola* in drinking-water.

1. General description

Fasciola hepatica (known as the common liver fluke) and *F. gigantica* (known as the giant liver fluke) are trematodes of the family Fasciolidae. Their natural hosts are mainly mid-sized and large-sized ruminants such as sheep, goats, cattle and buffalo. Pigs can be important reservoirs in human endemic areas. Humans can also be an important source of infectious *Fasciola*. Equines, including horses, donkeys, mules, and camelids, are alternative hosts (Mas-Coma, 2004).

Hybrid intermediate forms appear in areas of overlap of both species in Africa and Asia (Mas-Coma et al., 2009a). Adult *F. hepatica* flukes reach 29 mm in size and shed brownish-yellow, oval eggs with an evident operculum, measuring from 100 µm to 162 µm by 66 µm to 105 µm. Adult *F. gigantica* flukes reach 52 mm in size and with eggs measuring from 137 µm to 191 µm by 73 µm to 120 µm (Mas-Coma et al., 2014a).

Fasciola are vector-borne zoonotic organisms with complex life cycles involving several stages of development, including metacercariae, flukes, eggs, miracidia and cercariae and two-hosts (snails and humans) (Mas-Coma, 2004).

2. Human health effects

Four infection phases of fascioliasis can be distinguished in humans (Chen & Mott, 1990): i) the incubation period (from the ingestion of metacercariae to the first appearance of symptoms); ii) the acute, invasive or migratory phase (juvenile flukes enter the bile ducts); iii) the latent phase (fluke maturation and start of oviposition); and iv) the chronic, biliary or obstructive phase (fluke parasitism inside bile ducts).

The acute phase lasts from 3 to 4 months with a clinical picture that is mainly related to mechanical destruction of abdominal peritoneum and liver tissue (Acosta-Ferreira et al., 1979). Major symptoms include fever, abdominal pain (usually in the right upper quadrant), gastrointestinal disturbances, urticaria and respiratory symptoms. Signs of this phase include hepatomegaly and splenomegaly, ascites, anaemia, chest signs and jaundice.

The chronic phase may last from 9 to 13 years and comprise cholangitis and cholecystitis resulting from inflammation, epithelium hyperplasia, thickening and dilatation of bile ducts and gall bladder walls. Clinical manifestations include epigastric pain, fatty food intolerance, nausea, jaundice, pruritus, right upper-quadrant abdominal tenderness and biliary colic. Anaemia, lithiasis, cirrhosis, bleeding and multiple extrahepatic venous thrombosis may be further severe complications which can lead to death (Mas-Coma et al., 2014b). Sequelae may persist for years after treatment (Rondelaud et al., 2006). Juvenile flukes sometimes deviate during migration and may enter other organs and cause ectopic fascioliasis, which may be highly pathogenic or even fatal when affecting the central nervous system or eye (Mas-Coma et al., 2014b). The percentage of asymptomatic patients reported is 12% in developed countries (Arjona et al., 1995) and 15% in human hyperendemic areas of developing countries (Haseeb et al., 2003).

3. Epidemiology

Estimates of *Fasciola* infection range from 2.4 to 17 million people worldwide, while more than 51 million are at risk of infection (Mas-Coma et al., 2009a; Caravedo & Cabada, 2020). However, these estimates are dated, and given reports of expanding endemic areas, accurate updates to prevalence data are needed (Tidman et al., 2023). More recent analysis indicate that global prevalence could be higher than existing estimates (Infantes et al., 2023). Geographically, *F. hepatica* is distributed throughout Europe, northern and eastern Africa, Asia, Oceania, and the Americas. However, *F. gigantica* is restricted to Africa and southern Asia (Mas-Coma et al., 2009a; Tidman et al., 2023).

The regions with the highest prevalence of fascioliasis are South America (9%) and Africa (4.7%) (Infantes et al., 2023). Human endemic areas are known in Peru, Bolivia, Argentina, Chile, Venezuela, Mexico, Honduras and Cuba in the Americas. In Europe, human infection is higher in the western countries, mainly in France and Portugal and secondarily in Italy and Spain. In Africa, there are endemic areas in Egypt, Ethiopia, Yemen and Tanzania. In Asia, endemic areas include Türkiye, Iraq, Iran, Pakistan, Viet Nam, Lao People's Democratic Republic and southern China. In endemic areas, prevalence and intensity of infection peaks in 9- to 11-year-olds, but infection persists in adults (Esteban et al., 1999), with females appearing to be more affected (Mas-Coma et al., 2005). In high-income countries, the peak appears in 30- to 40-year-olds (Mas-Coma et al., 2014b). Infection is uncommon in North America, where most cases are associated with travellers and immigrants (Caravedo & Cabada, 2020). Within countries, human infection varies largely in relation to the distribution of snail vector populations in freshwater bodies and general physiographic and climatic conditions (Mas-Coma, 2005). Occurrence of fascioliasis is highly influenced by climate change and water resource development projects (e.g. construction of irrigation systems) and movement of animal reservoirs (e.g. livestock import/export, seasonal movement) (Mas-Coma et al., 2009a, b; Afshan et al., 2014).

Fascioliasis follows seasonal patterns influenced by temperature and rainfall, which vary depending on latitude and altitude and can include year-long transmission in zones where monthly temperature variations are small (e.g. northern Bolivian Altiplano); mono-seasonal transmission in extreme latitudes; the monsoon period in south central Asia and bi-seasonal transmission (spring and autumn) in areas of Europe, United States of America and Australia (Mas-Coma et al., 2018).

4. Transmission

Humans become infected by ingestion of metacercariae, the larval stage of *Fasciola*. Ingestion of wild freshwater plants such as watercress is the main source of infection. Drinking contaminated water and consumption of contaminated beverages, juices, and soups are receiving increasing attention as a source of fascioliasis. Washing of vegetables, fruits, tubers and kitchen utensils with contaminated water can also be sources of disease. The high diversity of infection sources and their heterogeneity in different countries underlie the large epidemiological heterogeneity of human fascioliasis (Mas-Coma et al., 2018).

Person-to-person transmission is not possible, as *Fasciola* only becomes infective in humans and animals after a four-phase life cycle that involves larval stage development in a lymnaeid snail host. In the first phase, the definitive host ingests metacercariae. The metacercariae excyst in the intestinal tract and migrate to the liver and bile ducts. After 3 to 4 months, the flukes attain sexual maturity and produce eggs, which are excreted into the bile and intestine. Adult flukes can live for 9 to 13 years in the host. In the second phase, the eggs are excreted by the infected host into fresh water where miracidia develop and are released. In the third phase, miracidia penetrate a snail host and develop into multiplicative larval stages producing cercariae, which are released into water. In the fourth phase, cercariae swim for a short period of time until they reach a suitable attachment site (aquatic plants), where they encyst to form metacercariae, which become infective within 24 hours (Mas-Coma, 2004). Metacercariae can also be found floating in water.

Fascioliasis is a typical vector-borne zoonotic disease. Ecology, behaviour and population dynamics of the freshwater lymnaeid snail vectors are the main factors which define the transmission patterns of disease (Mas-Coma, 2005). Small, cool temperature-preferring, amphibious lymnaeids of the *Galba/Fossaria* group which are distributed worldwide are the main vectors of *F. hepatica*. Larger, warm temperature-preferring and more aquatic lymnaeid species of the *Radix* group found in Africa and Asia are vectors of *F. gigantica*. These characteristics underlie the distribution of *F. hepatica* in colder countries and higher altitudes and *F. gigantica* in hotter countries and lowlands (Bargues et al., 2001).

The presence of livestock in or close to the household, the in-house use of water piped from canals, the habit of taking animals to the canals for bathing, and the use of animal manure on crops appear to be associated with *Fasciola* infection in humans (Curtale et al., 2003, Tidman et al., 2023). However, humans are also considered to be an important source of infectious *Fasciola* since high prevalence of fascioliasis in humans is not limited to areas where fascioliasis is a major veterinary problem (Mas-Coma, 2004; 2005). The numbers of eggs shed in human faeces depends on the number of adult flukes in the body; more than 8000 eggs/g have been reported in children in areas where human hyperendemicity is high (Mas-Coma et al., 2009a).

Evidence from an animal study indicates that metacercariae are highly infective¹, but the infective dose in humans has not been determined. In the study, rats became infected after receiving doses of 5 to 150 *F. hepatica* metacercariae with metacercariae age negatively influencing their infectivity. With doses of 20 metacercariae, the maximum longevity was 31 weeks (Valero & Mas-Coma, 2000).

5. Detection methods

Detection and identification of *Fasciola* is challenging and depends on microscopic detection of eggs in faecal samples from hosts by flotation and sedimentation methods followed by microscopic identification (Mas-Coma et al., 2014a) or in wastewater samples by centrifugal-

¹ From experiments with human volunteers, from epidemiological evidence and from experimental animal studies. Based on the criteria established by WHO in the Guidelines for drinking-water quality as follows: High means infective doses can be 1–10² organisms or particles, Moderate 10²–10⁴ and Low > 10⁴.

concentration and filtration methods (Hatam-Nahavandi et al., 2015). Contamination with metacercariae may be analysed by microscopic observation in plants and by filtration in water. Molecular methods are useful to confirm the classification of metacercariae as *Fasciola* but do not provide information about their viability (Mas-Coma et al., 2018).

6. Source and occurrence

6.1. Faecal pollution sources

Fasciola have been detected in wastewater containing slaughterhouse and livestock discharge, although in low proportions (5% to 7%) relative to other helminth eggs detected (Hatam-Nahavandi et al., 2015). In dairy farms in Colombia, *Fasciola* were detected in about 43% of wastewaters and about 54% of supply waters, which is thought to be due to the practice of using wastewater sludge as fertilizer in the area without stabilization or treatment (Rodríguez et al., 2012).

6.2. Surface water

Fasciola may occur in different types of freshwater (Mas-Coma et al., 2018). Temporary water bodies inhabited by lymnaeids explain links between seasonal transmission and rainy seasons. Permanent water bodies inhabited by lymnaeids in high altitude and high evapotranspiration areas allow for disease transmission throughout the year. Lymnaeids live in stagnant waters. Amphibious lymnaeids prefer puddles instead of deeper waters. The high colonization capacity of lymnaeids explains their presence in water bodies such as springs, artificial fountains and dams.

Metacercariae have a hard outer wall and can survive for long periods in water² but are susceptible to high temperatures and sunlight. In laboratory studies, *F. hepatica* metacercariae survived for up to 130 days at 10 °C, up to 36 days at 25 °C and up to 14 days at 30 °C (Boray & Enigk, 1964).

6.3. Groundwater

Fasciola are not found in groundwater since lymnaeids, their intermediate host, need light to survive.

6.4. Drinking-water

Reports on the prevalence of *Fasciola* in drinking-water are limited. Bargues et al. (1996) noted that in the Bolivian Altiplano, 13% of the metacercariae collected were from natural water used as a source of drinking-water.

² Based on the criteria established by WHO in the Guidelines for drinking-water quality as follows: Detection period for infective stage in water at 20 °C: Short, up to 1 week; Moderate, 1 week to 1 month; Long, over 1 month.

6.5. Soil and crops

Crops, such as lettuce, cultivated using water from irrigation systems inhabited by lymnaeid vectors and contaminated by livestock faeces, can be a source of metacercariae (Mas-Coma, 2004; Mas-Coma et al., 2018).

7. Significance of sanitation

Safe sanitation systems that effectively separate human faeces from the environment are an important barrier against contamination of water with *Fasciola* eggs excreted by infected humans. However, sanitation interventions may not be effective immediately due to the potential for long persistence in the environment (Boray & Enigk., 1964). Furthermore, as animals are the primary source of *Fasciola*, a comprehensive approach that deals with the management of animal faeces is necessary to control the risk of fascioliasis.

8. Significance in drinking-water

There is moderate evidence for *Fasciola* as a cause of waterborne disease. The importance of fascioliasis transmission through contaminated surface water used as the drinking-water source, is supported by indirect evidence (Mas-Coma et al., 2018). This includes three situations where statistically significant correlations of fascioliasis and other waterborne parasitic infections were demonstrated in human hyperendemic areas in Peru where people do not have a history of eating freshwater plants but collect water for drinking from irrigation channels inhabited by the snail vector *Galba truncatula*; in Bolivia where floating metacercariae or lymnaeid vectors are present in surface water collected for drinking; and a pronounced decrease in prevalence of fascioliasis through supply of filtered water primarily for washing clothes in communal village laundries, but also used as a source of drinking-water and for food preparation (Mas-Coma et al., 2018).

9. Prevention and management

Systematic approaches that identify and control faecal contamination along the whole sanitation service and drinking-water supply chains are needed to manage microbial risks, including from *Fasciola*. Since animals are the primary source of *Fasciola*, a comprehensive approach that also deals with the management of animal faeces is necessary to control transmission.

The effectiveness of sanitation and drinking-water treatment systems depends on the selection and operation of technologies and the characteristics of the target organisms. Treatment processes based on size exclusion (e.g. filtration through media or soil in septic tank drainage areas) or settling in primary wastewater treatment processes will reduce concentrations of helminths such as *Fasciola*. It is likely that oxidizing chemical disinfectants, such as chlorine will have limited effectiveness against *Fasciola*.

9.1. Sanitation systems

Sanitation safety plans (WHO, 2022) should include control measures to safely collect, treat and discharge or reuse wastewater and faecal sludges. Removal or inactivation of *Fasciola* by wastewater treatment processes will be variable and largely based on size exclusion. There are no reports on removal of *Fasciola* by on-site wastewater treatment processes.

9.2. Drinking-water supplies

Within a water safety plan (WHO, 2012; 2023) control measures to manage potential risks from *Fasciola* should include protection of source waters from human and animal faeces, application of adequate treatment and protection of storage and distribution systems from external contamination. Based on morphology and resistance to environmental pressures, metacercariae are likely to be relatively resistant to disinfection. However, based on their size (about 0.2 mm in diameter), they should be removed by physical processes. Sand filtration used to treat water supplied to laundries and used for drinking, food preparation and washing in Tiba, Egypt, reduced the prevalence of fascioliasis from 18% to 2% (Mas-Coma, 2004; Mas-Coma et al., 2018).

Due to the complex life cycle and environmental survival of *Fasciola*, *E. coli* (or, alternatively thermotolerant coliforms) is not a suitable indicator for the possible presence or absence of *Fasciola* in drinking-water.

10. ICD-11 classification

XN1H3	<i>Fasciola</i>
1F82	Fascioliasis

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Annex. Approach to content development and declarations of interest

The background documents were developed drawing from the following resources (i) microbial fact sheets from the fourth edition of the WHO Guidelines for Drinking-Water Quality (ii) pathogen-specific chapters in the Global Water Pathogen Project (GWPP) book “*Sanitation and Disease in the 21st Century*”, an update of the seminal book by Feachem et al. 1983 “*Sanitation and Disease: Health Aspects of Excreta and Wastewater Management*” and (iii) a rapid review of the literature at the start of the development of the background documents and before finalization in December 2023, focusing on identifying additional recent high quality literature reviews on a given pathogen. The documents were prepared by members of the WHO drinking-water quality microbiology group as well as external scientists with expertise in the relevant pathogens. The documents went through several iterations based on inputs from members of the WHO drinking-water quality microbiology working group and WHO technical staff. In addition, each document was reviewed by at least two external peer reviewers including authors of the GWPP book chapters. The extensive review process was designed to ensure technical accuracy and to identify any data gaps. The background documents were discussed at three expert meetings to resolve any issues and ensure the content reflected a consensus of expert opinions. Upon satisfactory resolution of any further data gaps or issues, the background documents went through technical editing before finalization.

Before the key expert meetings mentioned above, and for this work more broadly, all microbiology working group members submitted declarations of interest to WHO. These were to disclose any potential conflicts of interest that might affect, or might reasonably be perceived to affect, their objectivity and independence in relation to the subject matter of the guidance. WHO reviewed each of the declarations and concluded that none could give rise to a potential or reasonably perceived conflict of interest related to the subjects discussed at the meetings or covered by the guidance.